

## **Introduction**

Artificial Intelligence (AI) is the simulation of human intelligence by machines.

## **Machine Learning**

Machine Learning (ML) is a subset of AI that enables systems to learn from data without being explicitly programmed.

## **Deep Learning**

Deep Learning uses neural networks with multiple layers to identify complex patterns.

## **Natural Language Processing**

NLP enables computers to understand, interpret, and generate human language.

## **RAG**

Retrieval-Augmented Generation combines information retrieval with a Large Language Model to answer questions using external documents.

## **Vector Database**

A vector database stores embeddings and retrieves the most semantically similar chunks.

## **Embeddings**

Embeddings are numerical vector representations of text that capture semantic meaning.

## **Chunking**

Chunking splits large documents into smaller overlapping sections before creating embeddings.

## **Hallucination**

Hallucination occurs when an LLM generates information that is incorrect or unsupported by the provided context.

## **Conclusion**

A RAG system typically follows this pipeline: Upload PDF → Extract Text → Chunking → Embeddings → Vector Database → Retrieval → LLM Response.